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QUANTUM PSIONICS: A TIME-SYMMETRIC INTERPRETATION OF CONSCIOUSNESS

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ABSTRACT

In this thesis we will tie an interdisciplinary knot whose first rope is the Hard Problem of Consciousness. Having estabilished the nuances of the issue, we will explore the findings and consequences sprouting from Libet's 1983 paper, which insinuates Humankind has no such thing as free will, due to RP existing before the aware intention. Through bleeding edge findings in Quantum Mechanics, in particular those relative to superradiance, we will borrow Penrose and Hameroff's ORCH-OR to corroborate the idea that microtubules might be the seat of consciousness. Providing simulation-derived support to the theory that microtubules do in fact interact between each other via quantum processes linked to their tryptophan sites, we will then explore paradigm-breaking theories like Quantum Cosmology and Cramer's Transactional Interpretation of Quantum Mechanics. These Time-symmetric frameworks combined will allow us to produce a speculative but possible statement: it does not matter whether the RP comes before the thought, as both contribute to each other's existence, coexisting as the whole cosmo does in a tetra-dimensional block spacetime. As such, free will is framed not a mere desperate illusion anymore, but rather as an incredible feat of the universe itself.

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1. STATE OF THE ART AND INTRODUCTORY LITERATURE

1.1 The Hard Problem of Consciousness and its detractors.

Since well before the creation of the first psychology laboratory by W. M. Wundt, the widely recognized father of psychology (The Metaphysics Research Lab, Department of Philosophy, Stanford University, 2023), the would-be field of psychology and those who held an interest in comprehending the structure, functions, and dynamics of the human psyche found themselves at an impasse (Leibniz, G.W, 1714), centuries before the formalization of the field, when discussing one of the least understood psychic phenomena: consciousness (Chalmers, D.J, 1995). The vast and imposing topic of consciousness has allowed sufficiently expedite scientific advancement in some of its issues, mainly those that could be explained by finding their neural correlates or by the creation of a cognitive model (*ibidem*), for this reason called easy problems of consciousness. In contrast to this definition, the "really hard problem of consciousness is the problem of experience" (*ivi, pp. 203).*

To better comprehend the nuances of the hard problem, we need to explain what *qualia* are. When we feel a gust of wind on our skin or see the colors of dawn, in each of those sensory experiences there is a subjective quality, an introspectively accessible aspect (The Metaphysics Research Lab, Department of Philosophy, Stanford University, 2023) which is phenomenal in nature, i.e. it has the property of that which appears in our experience (The Metaphysics Research Lab, Department of Philosophy, Stanford University, 2023). The explanation of a *qualia* can't be reduced to the understanding of its underlying neural circuitry or of the way it's supposed to function via cognitive modeling. Subjective experiences thus elude any classical scientific explanation or method of enquiry, requiring a novel approach to underline what, how and why they are generated and rendered available to the psyche. Despite it's instinctually perceived importance, the shroud of mistery around consciousness is not universally considered an actual issue (Bourget, D. & David J. Chalmers, D.J, 2023), and some would argue that it's presence or lack thereof could be explained by solving all the easy problems that lead to the really hard one, effectively denying that the hard problem is in any fundamental way different from the easy problems (Dennet, D. C, 1996). Dennet (Dennet, D. C, 1995), in his brilliant and vitriol-filled comment on philosophical-zombies, argues that if a being behaves and, by extension, thinks (and feels!) like a conscious being, then it *does* have a consciousness and that the lack of it would render impossible the perfectly-humanlike behaviour of the zombie (*zimboes*, in his text). The author thus paints a very materialistic view of

consciousness, negating this statement: "...theirs are merely 'informationally sensitive', while ours are also 'experientially sensitive'. " (*ivi, pp.323).*

What appears clear in this exchange is that Dennet is reducing consciousness to a cognitive process, a function of the individual which expresses itself in behavior, thus discarding the issue regarding the presence or lack of qualia*,* treating them, at best, as byproducts of cognitive functions. This position by the author has already been expressed: "Qualia are supposed to be special properties, in some hard-to-define way. My claim… ...is that conscious experience has no properties that are special in any of the ways qualia have been supposed to be special." (Dennett, D. C, 1988). He then chooses to completely deny the existance of qualia as interpreted in their strictest form, with qualia answering to a principle of ineffability (*ivi, pp. 67*)*.* Dennet thus proposes his Multiple Drafts Model, stating that the psyche is composed of multiple parallel processes that interact with one another, producing the entirety of cognitive functions. By his view there's no separate entity named "consciousness", instead it is the result of the integration of said "drafts" into an apparently unitary narrative (Dennet, D. C, 1991). This reductionist, functionalist argument by Dennet tries to debunk the concept of qualia while actively bypassing their most defining characteristic: "not analyzable in terms of any explanatory system of functional states..." (Nagel, T, 1974), prompting the supporters of there being "...something that is like to be that organism..."(*ibidem*) to assertively contrast his ideas. Chalmers, contrary to Dennet's view, argues that subjective experience, in a theory of consciousness, should be considered a fundamental and irreducible element. Consciousness may arise from the material, albeit it wouldn't be entailed by it (Chalmers, D.J, 1995), achieving a non-reductive framework.

The hard problem is thus made explicit in all its non-triviality: if we are able to explain how DNA works regarding the inheritance of traits, we have explained its function and the function's performance, thus the concept. Consciousness, though, is not about the explanation of a function, rather of experience. we can't just hope to resolve this conundrum by merely defining "how information is discriminated, integrated, and reported…" (*ivi*, pp. 205), since it doesn't answer the question of how it is experienced. Why is it that when we perceive something we properly experience it, instead of it staying in the background, still performing its purpose as information digested and used, as an example, in navigating the environment? Why is it that we don't just catch a ball thrown at us, but we are aware, we experience the object flying towards our skulls, contrary

to what happens to people with blindsight (Weiskrantz, L. et al, 1974)? Blindsight, which allows people with damage to the primary visual cortex (V1) to respond to visual stimuli while reporting no conscious visual experience in part of their visual field, perfectly illustrates the difference between information processing, which here is completely intact and serviceable to other processes, and subjective experience, which here is instead completely lacking.

Supporters of the phenomenological view of the hard problem would interpret the aforementioned situation as proof that, in some yet-to-be-understood way, the qualia of the experience have been removed or rendered inaccessible to the conscious mind, thus in a way making it conceivable that it might indeed be possible to have a p-zombie, which can act exactly like a conscious being, albeit without the subjective character of the experience. Detractors of this interpretative framework would instead state that this holds no proof that there exists a consciousness which is holistic, monadic, and unfragmentable, but that instead all we have is a conglomerate of differently specialized neural circuits, an amalgam of narrative drafts that, in a case like that of blindsight, are partially or incorrectly integrated, denying that subjective experience is qualitatively different from the integration of information processes (Dennett, D. C, 1993). Going back to Dennet for a moment, he would explain the blindsight effect as nothing more than a failure by some informational draft to merge and integrate into the subject's unitary narrative. More sophisticated, specific theories, namely the Global Neuronal Workspace theory, state that neural networks are modular, active in parallel and allow for the unconscious processing of information. This information is then brought to consciousness when the neural module associated with it is attentively amplified into a state of coherent activity, a globally accessible workspace, via "...networks of neurons with long-range axons densely distributed in prefrontal, parieto-temporal and cingulate cortices." (Dehaene, S. & Changeau, J. P, 2011), which comprises the whole brain. This theory thus presents the idea that the ability of these neural modules to enact long-distance connections and therefore render the information available to many different processes is what we experience as a conscious state (Dehaene, S. & Naccache, L, 2001), and that in the case of blindsight, instead of lacking the qualia of subjective experience, the processes associated with vision are simply not able to be targeted by the attentive processes, twarthing the chances of the former processes of being amplified into a coherent, brain-wide state. In line with the opposition of there being qualia as separate, particular entites, the Global Neuronal Workspace theory therefore interprets the subjective conscious experience as the product of the globalization of a specific neuronal workspace, made possible by

the neuron's long-distance links, and not as a distinct, superimposed entity with its own independent or intrinsic properties, exempt from its underlying neural dynamics. Once it is explained which mechanisms make broadcast to the global neuronal workspace possible, following this point of view, no question about conciousness will be left unanswered. While the Global Neuronal Workspace theory associates the conscious experience with flexible control, error correction, inhibition, serial chaining, and strategic use of information (Marek, S. & Dosenbach, N. U. F, 2018), due to their workspaces' relation to fronto-parietal activity, there are authors which do not follow this strictly adaptive, performance-centered view of consciousness. The Higher-Order view, contrary to what Global Neuronal Workspace advocates, doesn't restrict or base the role of consciousness on performance enhancement. This is because the core of the theory argues that "...conscious awareness depends on higher-order mental representations that represent oneself as being in particular mental states" (Lau, H. & Rosenthal, D, 2011), and considering that most perceptual/cognitive tasks' performance depends on first-order representations, i.e. the base representations integrated by our perceptory systems (the shape and color of a ball), along with the fact that first-order representations are not always conscious, then one could agree with Higher-Order Theory when arguing that, for a representation to be brought into conscious awareness, it must become itself the target of a representation. The image painted by HOT is thus one of metacognition, of metarepresentation, stating that "...no mental state is conscious if one is wholly unaware of being in such mental state." (*ibidem*). When this theory is referred to when discussing blindsight, it can be said that the patients retain first-order (the base representation of the object) visual states which are sufficient to follow thourgh with action, but lack a representation of themselves being in that state, which explains the patient's lack of awareness. This paradigm proposed by Higher-Order Thought theory finds support in the fact that transcranial magnetic stimulation applied to dorsolateral prefrontal cortex impedes subjective reports of visual awareness, albeit without thwarting performance during tasks (Rounis, E. et al, 2010).

1.2 Replies to functionalist critics

Each of these theories in opposition to the hard problem share a functionalist outlook on the issue. While their contributions to the discourse are valuable, they ignore the issue at the heart of the problem untouched. Dennet's Multiple Drafts Model, Global Neuronal Workspace Theory and Higher-Order Thought theory all try to explain how and when information becomes conscious, accessible to the unitary narrative, but none explain why the availability of information to our cognitive network is accompained by conscious, subjective experience. As previously stated in this text, Dennet's proposal ignores qualia explicitely, relegating them to mere illusions of a consciousness that's, according to him, no more than the integration of its parts. Not quite a solution to the issue, rather an *a priori* refusal of the whole concept, which inadvertedly brings forth the concept of "explanatory gap". An explanatory gap, as proposed by Levine, is the fact that when we discuss things like heat, we can easily separate the subjective experience of heat from its physical mechanism (the molecules' vibration) and accept its mechanism as its identity, while if we try to do the same when discussing pain, then we encounter the issue that pain isn't easily discernible from the subjective experience to it associated. We could say that the feeling of pain is indeed pain itself: differently from before, we don't just gain access to the phenomenon through our perceptions, instead the experience of pain is exactly what we mean by "pain" (Levine, J, 1983). This makes it quite hard to reduce the subjective experience of pain to a physical mechanism, thus disallowing the explanation of the phenomenon via functionalist approaches. Global Neuronal Workspace Theory, on the other hand, links consciousness to the broadcast of a module to the workspace, but it still doesn't justify why the ignition should cause subjective experience and not only a greater integration of functions leading to a subsequent increase in performance. Higher-Order Thought theory gains reliability by correctly explaining why there can be a dissociation between a task being executed and the conscious report of it, but it still doesn't solve the hard problem. One could say that the theory solves reportability, since it's stated that a representation reaches consciousness when it itself is the target of an ulterior representation, but this still doesn't explain the subjective experiential quality, the phenomenology, the qualia pertaining to things like pain or colors. Some would argue that Higher-Order Thought theory is a reductionist one, interpreting consciousness as a mere cognitive monitoring and thus ignoring the intrinsic, eidetic qualities of subjective experience (Block, N, 2011).

1.3 Contextualizing the Hard Problem

This account of the many different interpretations of the hard problem of consciousness brings into the light a significant divide present between these diverse positions. Supporters of the functionalist approaches would be satisfied with the previous explanations, i. e. consciousness is nothing more than the integration of different functions, neural modules or the product of meta-representations. Such views would disintegrate the hard problem into nothing, denying it even existed in the first place, but in this paper we will entertain the more phenomenologically inclined approach, saying that none of these theories is capable of tackling the aspect of qualia, the intrinsic and seemingly ineffable quality of subjective experience. Therefore, when presented with the impossibility to provide a clarification of the concept of qualia in the current frameworks, we are pressed towards a change of paradigm. One of the proposed resolutions for the hard problem that doesn't fall into functionalist reductionism is Property Dualism, a consclusion readily reachable by following the discourse and refuting functionalist takes, states that consciousness is indeed related to physical states in the brain, but it itself is not physical, rather a fundamental element of reality that cannot be reduced to physical processes (Chalmers, D. J, 1996). Truly, this point of view seems to come up naturally and instinctively, radical only for the fact that we witness a standpoint explicitely rejecting materialistic views. Another solution proposed for the lack of explanations regarding qualia is Panpsychism, which argues that consciousness is not only a fundamental element of reality but that it is a fundamental feature of matter itself, assigning consciousness to even the more basic entities, like an electron, stating that even such a primordial particle may harbour some sort of proto-consciousness (Strawson, G, 2006). On almost the opposite side of the spectrum, Information Integration Theory states that, instead of having proto-consciousness in atoms, subjective experience exists as the product of integrated information, with a strong focus on what the whole system generates, which is, in perfect Gestalt fashion, more that the sum of its parts would suggest. In short, the amount of consciousness gained depends on the spread of integration, and the quality of consciousness (qualia) are the way these informational systems are wired together (Tononi, G, 2008). Thus, not something ineffable and mysterious, added on top of brain activity, rather consciousness is the integrated information itself, which, following this theory, could be found and measured even in non-biological informational systems, winking slightly at Panpsychism. The views opposite to the functionalistic approaches show us that it is indeed possible to tackle the hard problem without side-stepping the issues regarding qualia, all with varying degrees of success and positioned at different places on the spectrum that goes from materialism to more radical deviations,

like Property Dualism and Panpsychism. While they might disagree on most parts, all have in common that theories surrounding consciousness need more than a reductionist approach.

One theoretical framework yet to be discussed is an elegant, if ambitious, solution to the hard problem and more. This approach will be the core of this paper, tracing the path we might all choose to follow in search of a proper comprehension of consciousness and its quirks. Before introducing it, though, it's necessary to explain why such a particular approach is needed.

2. FREE WILL, CONSCIOUSNESS AND QUANTUM MECHANICS

2.1 Libet's proposal

The hard problem of consciousness, as previously stated and demonstrated, is an issue whose time dedicated on it spans centuries. Right beside the nature and function of consciousness, another daunting philosophical subject is that of *free will*. The papers and treaties that in time have tried to tackle this issue are innumerable (The Metaphysics Research Lab, Department of Philosophy, Stanford University, 2023), and the human mind often finds itself wondering, maybe more often in times of societal crisis, if the source of our very actions and thoughts is ourselves or if they're just a mere reaction to the environment. The difference highlighted by this issue is comparable to that found between a truly independent primate, choosing out of his own volition to either eat or sleep, to mate, to run or fight predators, and that of a leaf, now dry, waiting for the next gust of wind to lift it from the ground. How can the more introspective, reflective human beings sleep soundly when it's still debated if we are thinking beings, masters of our own bodies and minds, or in contrast we are just like boats that reached too far off the shore, tossed between the waves of a heartless chain of events and consequences?

To ease our minds, or to further curse us with a hardly quenchable thirst for closure, a scientist named Benjamin Libet and his colleagues chose to embark on a quest that would forever influence studies revolving around free will. In their paper, they have sought to measure precisely the temporal gap between brain activity and acts of conscious will (Libet, B. W. et al, 1983), discovering that the brain begins readying the firing of neurons relate to the execution of a voluntary act hundreds of milliseconds (circa 300, with varying standard deviation depending on the specifics of the action) before the subject becomes aware of the fact that they have chosen to act. This finding holds an immense weight: if truly our actions start to come into fruition before we are

aware of choosing to enact them, i. e. if neural initiation comes before conscious will, how can we still state that free will starts with a conscious decision? How can we say that it's our conscious, aware mind that chooses the next action? The implications of this study are that, instead, what we perceive as "free will" might be merely an illusion, in retrospect, rather than the source of our actions. An interesting particular of the findings is that, though the act is prepared before conscious awareness and thus conscious intention can't be the origin of such act, consciousness has still 150-200 milliseconds before the action is executed, and as such it could have some sort of corrective function, not dissimilar to what we see in the cerebellum regarding motor control and its associated strict time-control (De Zeeuw, C. I. & Ten Brinke, M. M, 2015). Though intrigueing, this stone consciousness might place on the paved road to action would still be susceptible to having an unconscious precursor.. This brings forth a point of view regarding free will that accepts consciousness not being the origin of conscious actions, but interprets it as a performance-control entity, in line with a compatibilist conception of free will (Libet, B. W, 1999).

Though later interpretations of Libet et al.'s findings state that the Readiness Potential (what was measured by Libet and colleagues and found to come before conscious awareness of the action/thought) might reflect more an accumulation of neural charge fluctuations waiting to breach the decision threshold, rather than a precise neural correlate of the decision process (Schurger, A. et al, 2012), the main issue arising from the paper's results remains untouched and unsolved: conscious decisive processes seem to be preceded by unconscious neural processes. This might not discourage compatibilists, accepting to reframe the concept of free will in a more regulatory role. For hard determinists the situation is quite different, as the fact that consciousness does not hold an ontological position in the decision making process signifies that free will as we usually conceive it is illusory (Lavazza, A, 2016), a product of the attempt of our minds to maintain a sense of agency. This gaping maw of a divide between conscious volition and the brain's preemptive activity causes an avalanche of following issues to discuss. If the concept of free will as we've always known it falters, how can we attribute moral and legal responsibilities to people and their actions? Libet's discoveries have set the stage which hosts a delicate induction: free will, if it exists at all, might not lie in the choice to act, but in the choice to interrupt an already initiated action, or change its course. Such an answer to this topic might not be welcomed with open arms by those who would prefer to refute such a restrictive view of their own, very personal agency. Maybe, a more radical change of

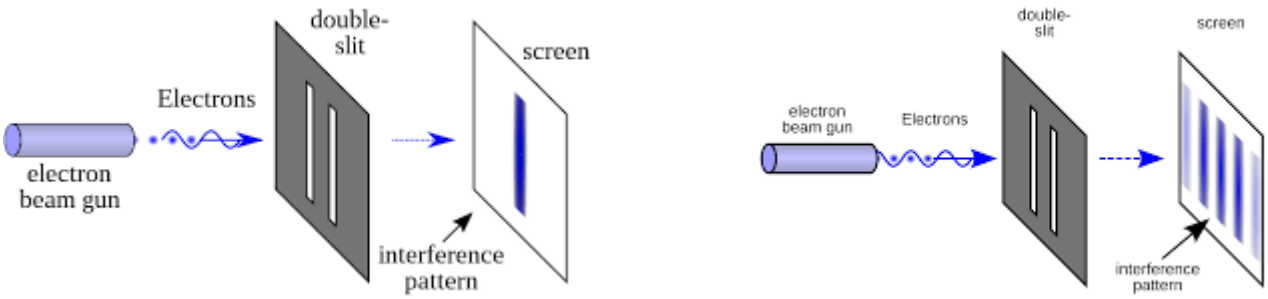
paradigm is required to crack this case, towards new answers and even more questions, in places and structures which are not classically associated with cognitive activity.

2.2 Neuronal microtubules and introductory QM

Since classical approaches seem unable to properly explore the hard problem of consciousness, the focus of this paper turns to a more peculiar paradigm. ORCH-OR, i.e. Orchestrated Objective Reduction, is a theory which states that consciousness resides not in the more classical places that are associated with cognition, but instead in deeper structures, called microtubules (Penrose, R. & Hameroff, S, 1996). Microtubules are a truly major and fundamental component of the neuronal cytoskeleton, sculpting and maintaining its structure which is based on the microtubule cytoskeleton itself (Kelliher, M. T., Saunders, H. A. J., and Wildonger, J, 2019), aiding and directing intracellular transport by, for instance, maintaining the correct orientation of their polarity (Jakobs, M. A. H., Zemel, A., Franze, K, 2022), depending on where in the neuron the microtubule is located, serving as some sort of rail to guide vescicles, organelles, proteins etc from the soma to the dendrites or the axon and vice-versa (Lasser, M., Tiber, J., and Lowery, L. A, 2018), help define which neurite (any part of the cell that protrudes from the soma (Flynn, K. C, 2013)) becomes dendrite or axon. Moreso, they are also capable of controlling crucial aspects of synaptic activity, like the release of neurotransmitters (Waites, C. Qu, X., and Bartolini, F, 2021), of activity-dependent polymerization and depolymerization, changing both their structure and function in the neuron (Peña-Ortega, F., Robles-Gómez, Á. A., and Xolalpa-Cueva, L, 2022), and of maintaining the cell stable by preserving structural integrity by responding to injury (Baas, P. W. et al, 2016). These remarkable structures have been proposed as a potential host of consciousness for the very specific reason that they might allow, though a series of qualities that will be explained shortly, quantum coherence to persist.

Before jumping into the theory's explanation, it would be most appropriate to first introduce the often intimidating topic of Quantum Mechanics. QM is the theory that describes the behaviour of matter and light at sub-microscopic scales (atomic and subatomic), where the physical systems exhibit qualities of both waves and particles (Feynman, Richard; Leighton, Robert; Sands, Matthew,

1964). This double nature of a subatomic particle/entity has been empirically confirmed via the so-called "Double slit experiment", the intricacies of which will be left to the reader, showing that when not being observed, photons exhibited a wave-like interference pattern when passing the two slits, but behaved like a particle if observed before entering the slits (Navarro, J, 2010).



Source: https://en.wikipedia.org/wiki/Double-slit_experiment

Now, such a property defies any and all classical theories of physics, and as such needs something that captures both of its natures. Here's where one of the most well-known aspects of QM shall be introduced[1]:

Schrödinger's Equation:

$$i\hbar\frac{\partial}{\partial t}\Psi(x,t) = \left[-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x,t)\right]\Psi(x,t).$$

This partial differential equation describes how the wave-like aspect of the quantum system, called the wavefunction, evolves in time and determines observable (particle-like) outcomes, in systems that do not come close to the speed of causality (non-relativistic quantum mechanical systems) (Feynman, Richard; Leighton, Robert; Sands, Matthew, 1964). The wavefunction, usually represented with the greek letter psi (**Ψ**), is obtained by solving the Schrödinger Equation, but how can a function of *x* for any given time *t* describe the state of a particle, if the wavefunction is by definition spread out in space? To solve this issue, **Born's statistical interpretation** tells us that the squared mode of the wavefunction derived in x indicates the probability of finding the particle in a range of space at time t:

$|\Psi(x,t)|^2\,dx$ = probability of finding the particle between *x* and *(x+dx)*, at time *t.*

1 Standard quantum-mechanical expressions, formulas *et simila,* otherwise specified, are to follow the core text from Griffiths and Schroeter (2018)

Truly, this perfectly introduces the concept of indeterminacy in QM, as we only have a probability of finding a particle, not a certainty. What does this mean then, subscribing to the **Copenhagen interpretation**, is that the observation itself causes the particle to collapse its multitude of superpositioned states into a definite, irreversible state (*ibidem*, as will be until further sources appear).

2.3 Understanding Quantum Mechanics

The wavefunction is thus involved in two processes: the first is an ordinary evolution, the second is the measurement process, in which Ψ collapses suddendly. To expand on this it must be said that the wavefunction Ψ is a mathematical object which holds all the information that can be extracted from a quantum system, and for this information to be obtained and manipulated the wavefunction must live in a Hilbert space, i. e. an inner product vector space that is also a complete metric space (Debnath, L. & Mikusinski, P, 2005). This allows the use of **linear operators** on the wavefunction, which is the way physical quantities are represented when acting on a Hilbert space, which in turn allows the use of linear algebra, following QM rules, to, as an example, discover the energy of a quantum system by applying the Hamiltonian operator $\hat{H}$ (which represents the total energy), whose eigenvalues give us the measurable results, while Ψ(x,t) gives us the coordinate representation of the state vector (the mathematical object representing a state of the system, indicated with Dirac's notation as |Ψ>). As an example, to compute the inner product of two distinct quantum states Ψ and Φ we would have to write it like this:

$$\langle \Psi | \Phi \rangle = \int_{\mathbb{R}^3} \Psi*(x)\Phi(x)\,d\mu(x)$$

Which is read: the inner product between quantum states psi and phi is equal to the integral over all space of the complex conjugate of psi multiplied by phi, integrated using the Lebesgue measure, to highlight the fact that we're in a Hilbert space. To further familiarize ourselves with the bra-ket or Dirac's notation, along with stabilizing the concept of a vector state, the wavefunction psi can also be written like this:

$$\Psi(x,t) = \langle x | \Psi(t) \rangle$$

Here, the wavefunction Ψ(x,t) provides the coordinate representation of the abstract (as it has no fixed coordinate system) state vector |Ψ(t)> in the position basis, allowing for the application of Born's statistical interpretation. This permits us to obtain the probability density of the quantum system within a given range of positions at a time t by taking the integral over all space ($\mathbb{R}^3$) of the

squared modulus of the wavefunction $|\Psi(x,t)|^2$. The probability density, which satisfies the normalization condition when integrated over all space is (*ibidem*):

$$\int_{\mathbb{R}^3} |\Psi(x,t)|^2 = 1$$

The quantum systems here presented all have quantized energy packets, as it will appear obvious when solving a Schrödinger's Equation, and from the solution to the SE, we can obtain information related to the operators or observables applied to the wavefunction of the quantum entity. An observable is a **Hermitian operator**, which means it's an operator that is equal to its transposed conjugate, written as $\hat{H} = H^{\dagger}$, and observables, as said before, correspond to measurable quantities, since their eigenvalues are real (Miller, David A. B, 2008). However, not all observables can be known precisely at the same time, due to some operators being non-commuting, because in QM the order in which two operators act on a state vector matters. This formula : " $[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A} \neq 0$" reads as "the commutator of observables $\hat{A}$ and $\hat{B}$ is not equal to zero", which means that the order in which these operators act on a state is important and that it's impossible to obtain precise information on both of them at the same time. From the concept of non-commuting operators we reach one of the more interesting aspects of Quantum Mechanics: **Heisenberg's Uncertainty Principle.**

Taking into account the expectation values of the non-commuting Hermitian operators $\hat{A}$ and $\hat{B}$ made being:

$$\langle A \rangle = \langle \psi | \hat{A} | \psi \rangle, \qquad \langle B \rangle = \langle \psi | \hat{B} | \psi \rangle.$$

we define **deviation operators**, mathematical objects which represent how much the observables A and B, again represented by the Hermitian operators $\hat{A}$ and $\hat{B}$, deviate from their mean values when in the chosen state:

$$\Delta \hat{A} = \hat{A} - \langle A \rangle, \qquad \Delta \hat{B} = \hat{B} - \langle B \rangle.$$

The expectation value of both deviation operators is zero

$$\langle \psi | \Delta \hat{A} | \psi \rangle = 0, \qquad \langle \psi | \Delta \hat{B} | \psi \rangle = 0.$$

From these, we can compute the standard deviation of the observable in a given state, which is the uncertainty in the measurement of each observable when the system is in a given quantum state:

$$(\Delta A)^2 = \langle \psi | (\Delta \hat{A})^2 | \psi \rangle, \qquad (\Delta B)^2 = \langle \psi | (\Delta \hat{B})^2 | \psi \rangle.$$

The relation between the uncertainties of two non-commuting observables is called **Heisenberg's Uncertainty Principle** and states that**:**

$$\Delta A \Delta B \geq \frac{1}{2} |\langle [\hat{A}, \hat{B}] \rangle|.$$

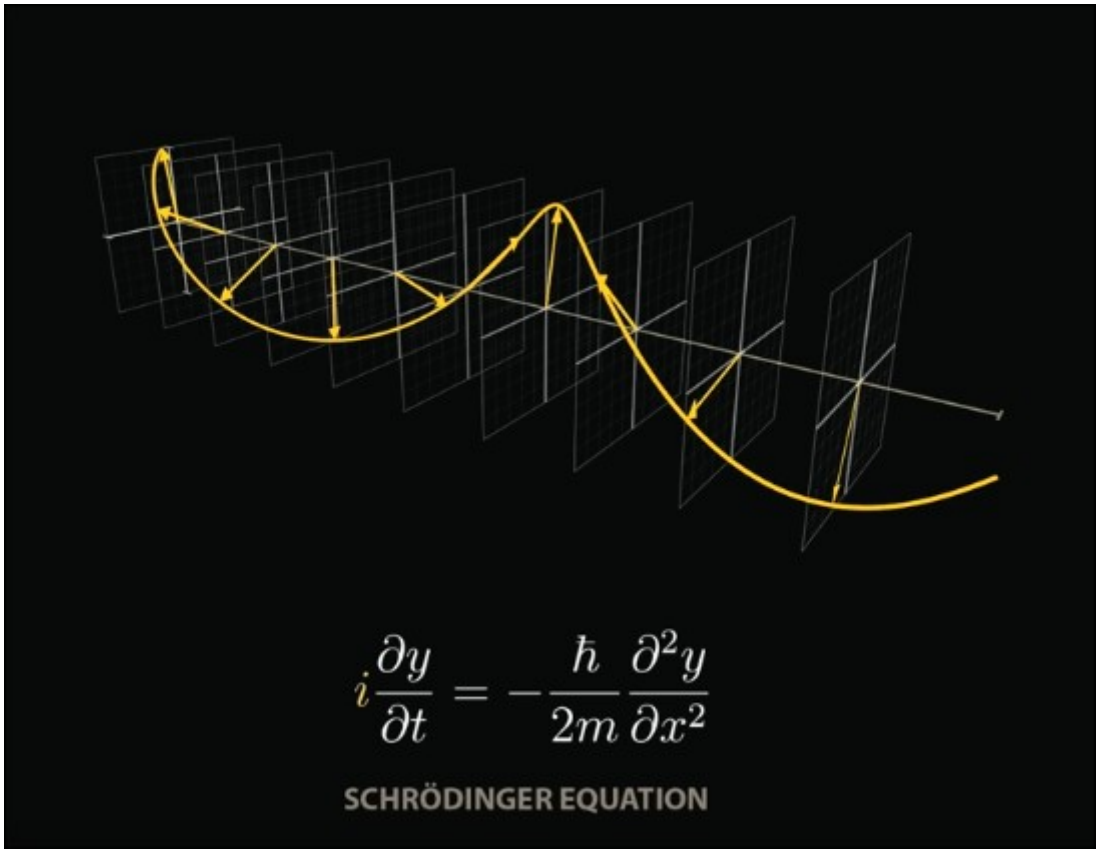
Meaning that the product of the two uncertainties is tied to the expectation value of their commutator, so that the more non-commuting are the observables, the higher is their minimum uncertainty between themselves. Famously, the position-momentum pair of observables have their uncertainty relation bound by this formula:

$$\Delta x \Delta p \geq \frac{\hbar}{2}$$

Demonstrating that the more we can extract precise information about the position of a quantum system, the less we can know about its momentum and vice versa, not because of limited measurement capabilities, but due to the nature of the wavefunction itself and of the matematical relationships between position and momentum spaces, both **Fourier Transforms** of each other:

$$\phi(p) = \frac{1}{\sqrt{2\pi\hbar}} \int e^{\frac{-ipx}{\hbar}} \psi(x) dx.$$

In Fourier analysis, the narrower a function is in one space, the broader it is in the other, meaning that if we're able to extract precisely the information about position, thus having a small Δx for our precisely localized $\psi(x)$, then applying the transform to reach the momentum basis $\phi(p)$ will yield a large Δp, which is exactly what's stated by Heisenberg's Uncertainty Principle. Another fundamental quality of the wavefunction is its **Phase.** The phase of a wavefunction is the offset angular position, the initial angular displacement between one oscillation and another taken as reference. In the complex plane, we can represent the wavefunction as a phasor, a rotating vector, and the phase is its angular coordinate. We can thus say that the phase represents how far the oscillation has progressed within a period, a way to measure the wave's rhythm and its rhythm relative to another wave. Just below this paragraph, a visual representation of what phase is:



source: https://www.youtube.com/watch?v=uVKMY-WTrVo

The yellow arrows in this image represent the phase rotation of the wavefunction, as it evolves in the complex plane. The curve is the real part of the wavefunction $\psi(x,t)$, its magnitude, while the angle or direction of the arrow is the phase $\phi(x,t)$. From the concept of phase we can arrive at one of the phenomena that stands closer to what the simulation will ask of us, while also being fundamental to the comprehension of QM: Quantum Coherence.

Quantum Coherence happens when the relative phases of the superpositioned components of a quantum system are stable. To better explain, consider:

$$|\psi\rangle = c_1|1\rangle + c_2|2\rangle$$

This is a quantum system which exists in a superposition of states 1 and 2, where $|1\rangle$ and $|2\rangle$ are the basis states and c_1 and c_2 are the complex probabilitity amplitude whose relative phase $\Delta\phi = \theta_2 - \theta_1$, more precisely: $\Delta\phi = arg(c_2) - arg(c_1)$, when stable in time, determines the **coherence** between the components. Coherence allows the quantum system to exhibit its quantum properties, as coherence allows for phenomena such as **interference effects**. Interference permits the addition

of not only the probabilities of the quantum states, but of their amplitudes, leading to **interference patterns** both constructive and destructive. The interference effects, both for a single quantum system or multiple, thus, let the system(s) maintain its wave-like qualities, its non-classical behaviour and allows more systems to collectively interact with each other.

2.4 Interference effects and non-classical behaviour

Let us take into consideration a probability density for a certain quantum system with two superpositioned states as:

$$\rho = |\psi|^2 = |c_1|^2 + |c_2|^2 + 2\,Re(c_1 + c_2^*)$$

Here, the measurable probability density is the square modulus of their sum, not simply the sum of the components' probabilities, and the last term $2Re(c_1+c_2^*)$ is the **interference term**. This mathematically shows the wave-like behavior of the system, as that interference term will produce peaks when the phases between the two superpositioned states' amplitudes are aligned, but will cancel out when they are opposite, leading to valleys in the graph, signifying zones of low to zero probability densities. This alternating rhythm of the interference pattern is typical wave behaviour, further confirming that it's senseless to consider quantum systems classically, as they possess an oscillatory, delocalized nature which translates to a continuous probability distribution dependent on the relative phases. By contrast and presented here only for completeness, the classical way of adding independent probabilities is $P = P_1 + P_2$, where both probabilities are indipendent, don't reside in a complex vector space and thus are unable to influence each other directly, lacking any capability for interference, be it constructive or destructive, exposing their nature as mutually exclusive events. Coherence, as fundamental a feature as it is, is quite sensible and fragile. Quantum properties are not easily maintained in a noisy environment, as anything and everything will disturb the quantum systems. When this happens, the delicate phase-equilibrium of the superpositioned states can be lost, in a process called **Decoherence**. During this phenomenon, the quantum system becomes **entangled** with its environment, which means that from that point on, the two components of the entanglement, i.e. the environment and the quantum system itself, can be described with a single wavefunction. This signifies that both components now cannot be described individually, as seen in this examples with the entangled particles A and B:

$$|\Psi\rangle = c_1|0_A 1_B\rangle + c_2|1_A 0_B\rangle \ .$$

The wavefunction exposes the fact that neither particle possesses an independent state, meaning that, no matter the distance, measurement of either particle instantaneously determines the other, a phenomenon famously called "*Spooky action at a distance*". Noting that the previous formula is very similar to $|\psi\rangle = c_1|1\rangle + c_2|2\rangle$ leads us to consider that the probability density for this entangled system will also be $\rho = |\psi|^2 = |c_1|^2 + |c_2|^2 + 2\,Re(c_1 + c_2^*)$. This shows that, eidetically, **entanglement is coherence across systems**, due to the relative phase of the components allowing, if stable, quantum effects like interference. Unfortunately, the ability of quantum systems to achieve coherence and thus entanglement is also the reason for their vulnerability, due to the disruption of phase relations. The resulting system-environment wavefunction is this:

$$|\Psi_{total}\rangle = \sum_i c_i |s_i\rangle_{system} \otimes |E_i\rangle_{environment}$$

Which reads as "the wavefunction is now comprised of the tensor product of both system and environment state vectors", where every environmental state $|E_i\rangle$ becomes correlated with a particular system state $|s_i\rangle$, as an example visualized like this:

$$|\Psi_{total}\rangle = c_1|1\rangle|E_1\rangle + c_2|2\rangle|E_2\rangle$$

Where we have the wavefunction comprised of particle/system basis states 1 and 2 correlated to the environment states E_1 and E_2, a composite system that, if we were certain it was closed, we would describe it as done just earlier, but since we assume it's an open system, it can be described as a **density operator**, basically "opening" the matrix by doing the **outer product** of their combined wavefunction:

$$\rho_{total} = |\Psi_{total}\rangle\langle\Psi_{total}|$$

Entanglement prohibits the association of a state vector to a corresponding subsystem, since now that both the Environment state vector and the System state vector are entangled, they **both** contribute to the wavefunction, and singling out one or the other wouldn't yield a wavefunction as a result, because now their vector states are mixed, but by describing it with a density operator we can apply a partial trace to filter out the environment, obtaining the object ρ_S which encodes all the information about the system by itself, ready for further manipulation: $\rho_S = Tr_E(\rho_{SE})$.

Since further chapters of this thesis will delve heavily in the use of density operators to extract information from multiple correlated quantum systems, and to further clarify why exactly two entangled systems lose the ability of being described by a pure state vector, we will now explore the **partial trace of a density operator**. Consider:

$$\rho_S = Tr_E(\rho_{SE}) = \sum_{v=0}^{M} (I_A \otimes X_v|)\rho_{SE}(I_A \otimes |X_v\rangle),$$

where M is m^{2-1}, m being the number of basis states of the subsystem Environment. Said Environment subsystem, when applying partial trace, is being averaged over all its possible states in order to access the information related to the System subsystem, which, if the Environment states are distinguishable and thus orthogonal, will reveal the absence of its off-diagonal terms when looking at its density operator ρ_S in matrix form. Mathematically, it can be stated that, if:

$$\langle E_2|E_1\rangle = 0,$$

then the resulting density operator:

$$\rho_S = |c_1|^2|1\rangle\langle 1| + |c_2|^2|2\rangle\langle 2| + c_1 c_2^*\langle E_2|E_1\rangle|1\rangle\langle 2| + c_1^* c_2\langle E_1|E_2\rangle|2\rangle\langle 1|,$$

shown in matrix form as:

$$\rho_S = \begin{pmatrix} |c_1|^2 & c_1 c_2^*\langle E_2|E_1\rangle \\ c_1^* c_2\langle E_1|E_2\rangle & |c_2|^2 \end{pmatrix},$$

when reduced due to the E states' assumed orthogonality amounts to:

$$\rho_S = \begin{pmatrix} |c_1|^2 & 0 \\ 0 & |c_2|^2 \end{pmatrix},$$

showing that subsystem S' states hold no coherence (the off-diagonal terms) between themselves, resulting in something that looks like a classical system, since now, again, the quantum properties are attributed to the composite system $|\Psi_{SE}\rangle$ and are not present in the single subsystems when isolated. The result instead is a classical probability distribution.

What happened here, in the moment the subsystem lost its off-diagonal terms due to entanglement with external systems is called **Decoherence** (Zurek, Wojciech, H, 2001). While the composite system can be described using a wavefunction, having a coherent quantum state, the subsystems cannot, because their phase relationships (the off-diagonals) are now lost to us, as they are correlated with the environment, making the subsystem appear classical (it does behave classically, but it's still part of the quantum system when said system is viewed in its totality). This phenomenon holds particular relevance in the theory of quantum cosmology, where the universe itself is considered quantum, treated as a wavefunction, but the entanglement effects present make normal spacetime appear classical (Kiefer, Claus, 2012). Having roughly understood what entanglement and decoherence effects are, we are now ready to approach quantum mechanics in

less than favorable environment, such as live tissue, to then better explore the role of microtubules in this here discourse about free will.

3. Quantum Mechanics in Biological Systems

3.1 Early proposals for Coherence in biological systems

The presence of Quantum Coherence effects in biological processes has been speculated since the late 1960s, when it has been proposed that biological systems, though not often respecting the notion of spatial order typical of crystalline structures, do instead subscribe to different kinds of order. One such type of order describes how particles may move and interact, in a coordinated and **phase-correlated** way, the so called motion or phase order, present in systems that can be described with a single wavefunction, such as superconductors or superfluid, allowing for the description of a system containing trillions of particles with a "macroscopic" wave function (Fröhlich, H, 1977). This proposal bases its hypothesis on dipole oscillators, such as membrane proteins or cytoskeletal polymers, which, when receiving more metabolic energy than a certain threshold value, tend to push most energy channels (in the article that's how a mode of the oscillators is called) towards the lowest-frequency mode, in a way analogous to what happens in Bose-Einstein Condensates (*ibidem*). In the author's own words: "When the supplied energy exceeds a certain value, a steady state develops in which the lowest-frequency mode contains an enegry very much larger than that of all others. This represents a kind of coherent excitation of the system" (*ivi, p.404*), stating that when the dipoles are In phase, oscillating together, they settle to a low-frequency "coherence mode", otherwise when they are out of phase they produce higher frequency modes. The author thus suggests that this coherent excitation of the dipole modes might provide some answers on how biological systems are organized. Following up on Fröhlich's proposal, Quantum ElectroDynamics (QED) has been used to describe the behaviour of a collective composed by water and other biomolecules. The authors state that water molecules can achieve coherence through coupling between molecular dipoles and the elctromagnetic field, sanctioning the emergence of a long-range order, a reduced entropy. To support their claim, the authors linked the behaviour of water domains to that of Free Electron Lasers (Del Giudice E., Preparata G., Vitiello G, 1988). FELs work by making the electrons oscillate transversely to their direction of motion via an undulator's periodic magnetic field. These electrons' motion induces an oscillating **electric dipole moment,** a concept fundamental to the explanation of this thesis' conceptual superradiance simulation which will be more thoroughly explored in later chapters of this thesis**.** An electric dipole moment is the measure

of the electrical polarity of a system of charges, i.e. a measure of the separation of charges inside the dipole, formalized as $\rho=\sum_i q_i \boldsymbol{r_i}$ for a system of point charges, where q_i are the magnitudes of the charges and r_i are vectors of distance between the charges. Realistically, in a molecule many charges aren't paired into +q and -q, so the dipole moment will be intended as a property of the whole charge distribution: $P=\int \boldsymbol{r}\,\rho(\boldsymbol{r})\ d^3r$, giving us the vector that points from the center of the negative charge to that of the positive charge, considering $\rho(\boldsymbol{r})d^3r$ the total charge in a differential volume element d^3r situated around **r**, adjusted based on the amount of charge. An **oscillating electric dipole moment**, conversely, requires us to account for the change in time of the charge separation: $\boldsymbol{P}(t)=\boldsymbol{P_0}\cos(\omega t)$, with $\mathbf{P}_0$ being the amplitude and ω as the angular frequency. In quantum mechanics, a **transition dipole moment** is instead associated to each transition between quantum states, formalized as $\boldsymbol{P_{eg}}=\langle e|\hat{\boldsymbol{P}}|g\rangle$, or in complex form as $\boldsymbol{P}(t)=\boldsymbol{P_0}e^{-i\omega t}$, which looks very similar to the time evolution of a wave state: $\psi(t)=\psi(0)e^{-iEt/\hbar}$.

After the induction by the FEL's undulator of the oscillating dipole moment, electrons interact with the electromagnetic field via phase-locking, leading to a diffused coherent mode in the electron beam and an amplified magnitude of emitted radiation (*ibidem*). In their QED model of water the authors propose that when molecular water dipoles become phase locked, they then interact with the electromagnetic field, reinforcing the coherence obtained by the water molecules. Where water molecules behave as a single entity, there we have a **coherent domain**, resembling how the previously mentioned FELs achieve self-reinforcing coherent states. This similarity between these two phenomenon leads the authorst to then state that this type of self-organizing systems might provide some answers on how biological systems maintain order and efficiency (*ivi, pp.*). These early attempts to trace a link between quantum mechanics and live, biological systems, though speculative and quite difficult to test in an experimental environment, acted as a basis for a conceptual framework that now included live tissue and biological environments as well as quantum mechanics.

3.2 Microtubules and Quantum Coherence

Microtubules are remarkable structures. As previously stated, not only are they fundamental to the cytoskeleton's architecture and for the neurons' intracellular transport system, but are also surprisingly regular in both their structure and electronic organization, making them promising candidates for Quantum Coherence. Microtubules are composed of tubulin dimers of α and β tubulin, containing several amino acids, in particular tryptophan sites. The π electrons of these tryptophan molecules are delocalized, possessing enough dipole moments to achieve coupling with the electromagnetic field, exactly as seen with FELs and coherent water domains. Since these dipoles are located in the tryptophan sites and they themselves are in the (quasi) periodic microtubule lattice, they can, in theory, lead to collective excitations due to their continuous electromagnetic interactions. These most peculiar characteristics led Hameroff's to consider them more than just entities dedicated to the maintainment of cytoskeletal structure and the facilitation of intracellular transport, and instead suspecting that they might have a direct role in neuronal information processing (Hameroff, S. R, 1987). Tubulin dipoles, he proposed, could achieve synchronized oscillations when influenced by metabolic energy, and if sustained these coherent excitations would allow microtubules to function as information-processing networks within the neuron itself.

Later, the idea was expanded upon, birthing the theory now known as **ORCH-OR**. The Orchestrated Objective Reduction theory proposes a model of consciousness where tubulin dimers are interpreted as quantum systems with two states which can exist in a superposition of conformational (spatial arrangement that can be changed by rotating around single bonds) or electronic configurations and each different conformation of the molecule is associated with its own charge distributions and dipole orientations inside the tubulin dimer. The two conformations of a tubulin dimer can be represented as a qubit with two basis states $|0\rangle$ and $|1\rangle$, written as the superposition $a|0\rangle+b|1\rangle$, where a and b are the respective complex probability amplitudes. Due to the fact that we find tubulin dimers in a microtubule lattice, they are spatially organized, and when many are found close to each other, their dipoles interact and couple (called **dipole-dipole coupling**). These interactions allow **excitations**, i.e. changes in dipole energy or orientation, that can propagate through the lattice in a collective manner. This system can, in theory, support collective quantum states, in a way not dissimilar to what we have seen before when mentioning FELs and coherent water domains. The theory states that networks of tubulin dimers (and thus microtubules), with their interacting dipoles, could form some sort of **quantum computer** in the

neuron, which would store and elaborate information through superpositions and entangled states, painting the image of a truly delocalized and interconnected system, which would in theory allow for much faster information exchange inside the neuron, due to the fact that a qubit can represent both 0 and 1 simultaneously, permitting parallel computation, contrarily to what classical computers can achieve. This non-classical way of computation would reflect the perceived integrated activity of the brain. The microtubules, composed of segments each made of 10^5-10^6 dimers, are interconnected throughout the neuron by the cytoskeleton, so their collective quantum behaviour could influence large-scale processes inside the neuron, working as a quantum, sub-cellular information processing unit.

3.3 Objective Reduction

Now that we have estabilished microtubules as vessels for quantum coherence in the neuron, we might ask ourselves how and why these coherent quantum states collapse, and, most importantly, how their collapse might correspond, subscribing to what the ORCH-OR theory states, to a **moment of consciousness**. Classical Quantum Mechanics attributes the collapse to observation or measurement, leaving the question unanswered. From this perspective, if the tubulin dimers' dipoles do couple and allow for quantum states which are maintained in time, they still will collapse, in order to produce definite outcomes in the neuron. While classical QM does not provide a true answer, the ORCH-OR theory associates each collapse of a quantum state of the tubulin dimers' network to a discrete gravitational event. Penrose refuses attributing to decoherence the collapse of the wavefunction. Instead, he chooses to propose a more **objective** reduction of ψ. Suppose we have a quantum system of superpositioned state, each with their own **mass distribution** $\rho_i(\boldsymbol{r})$ (not to be confused with the density operator), which is a way to represent a continuous mass density field in 3d space. The superpositioned states can be represented as $|\psi\rangle = a|1\rangle + b|2\rangle$ and their mass distributions produce two different gravitational fields, which means they are also two superpositioned space-time curvatures. Penrose suggests that this configuration is unstable. Let's now entail the difference of mass distributions $\Delta\rho = \rho_1(\boldsymbol{r}) - \rho_2(\boldsymbol{r})$, which we need to then measure the **gravitational energy** of the difference. One (the author of this thesis) would assume that the chosen gravitational framework would be the widely recognized General Relativity, but trying to insert GR into Penrose's theory breaks the model instantly, as if it was a syntax error in a python script. That is, General Relativity has no concept of "gravitational energy". The curvature in a

region of spacetime is both gravity and the position. Considering two superpositioned quantum states, if we subcribe to the point of view that the particle truly is in superposition and it's not a measurement issue or a lack of tools that forbids us to get rid of the probability and find a certain position, momentum, etc, then the superpositioned states correspond to two different configurations of spacetime geometries, which are **not** comparable, as there is no overarching reference configuration of a "base universe". This already leaves us with the impossibility of calculating gravitational differences, be they of configuration or gravitational energy, as we are left with two completely alien universes, absolutely not comparable. The more intricate and maybe interesting issue comes from the fact that, in GR, gravity isn't a force or a field over a base spacetime, it is instead spacetime itself, its curvature. The equivalence principle always allows us to consider a local frame where gravitational force vanishes (locally), like a free-falling elevator, leading to a Minkowski spacetime. This means that neither gravitational energy nor its differences can be computed, since any local gravitational field can be annulled by changing the coordinates. These issues are the reason why Penrose chose to compute its gravitational collapse using newtonian gravity, which assumes a standard background. The expression used to compute the **gravitational self-energy of the difference between mass distributions of superpositioned states,** i.e. the gravitational tension between two or more different spacetime geometries, is as such:

$$E_G = \frac{-G}{2} \int \int \frac{\Delta\rho(\boldsymbol{r})\Delta\rho(\boldsymbol{r}')}{|\boldsymbol{r}-\boldsymbol{r}'|} d^3r\, d^3r'.$$

This equation computes the gravitational self-energy associated with the difference between two mass distributions in a superposition, integrating over all pairs of points (G/2 is there to avoid counting twice) to determine how much the two configurations of spacetime created by the mass distributions exhibit gravitational differences. Following Penrose's train of thought, it measures the gravitational instability of the superposition, thing that will inevitably lead to collapse if the differences are too great, according to the **OR timescale:** $\boldsymbol{\tau} \approx \frac{\hbar}{E_G}$. The lesser the gravitational instability, the longer the superposition resists before collapsing. Hameroff then closes the deal proposing that the required number of tubulin dimers in superposition to reach a sufficiently large $\boldsymbol{E_G}$, in order to achieve the collapse , considering $E_G^{total} \approx N E_G^{(1)}$ as a collective OR collapse of N tubulins, to be such that $\boldsymbol{\tau} = \frac{\hbar}{E_G} \approx 25\,ms$, a frequency that matches gamma wave synchrony rythms that are associated with conscious awareness. Each ORCH-OR event is then a moment of conscious

awareness, and their venue, according to the authors, might be pyramidal neurons due to their tight bundling of long microtubules, which **could support collective dipole oscillations**, stabilized by Microtubule Associated Proteins.

Even so, this and other more or less classical interpretations fail to justify the existence of free will. Truly, most interpretations of both consciousness and the universe itself fail to provide a satisfactory answer to this deterministic *vulnus*. Even with microtubules as the seat of consciousness, we still do not resolve the time discrepancy between RP and conscious awareness of intention, surrendering to the fact that some kind of electrical activity comes before our awareness on the matter. But what if the key to solve this conundrum lies not within space, energy or matter, but instead within the dimension of time? Although Orch-OR proposes a mechanism for the emergence of single moments of aware consciousness, it doesn't address pre-conscious activity at all, leaving us with a yet still deterministic view of free will: its absence. For this reason, a shift in paradigm is needed, leading us to peculiar interpretations of both quantum mechanics and the universe at large.

3.4 Transactional Interpretation of Quantum Mechanics

The Copenhagen Interpretation states that, while the Schrödinger Equation describes the continuous evolution of the quantum system, which is mathematically time-reversible , collpase is instead a unitary, discrete and irreversible process which is triggered by measuring the system and is **not** derived from the SE. This implies that the SE does not encode all the information required to predict the outcome of the evolution of the quantum system, without providing a clear picture of what exactly happens when the system collapses and bringing in a time-asymmetry where in origin there was none. In truth, most fundamental equations in physics care not in which direction time flows, evident with the SE itself as changing t to -t, also taking the complex conjugate of the wavefunction, produces a valid solution. Mathematically, nothing forces it to proceed only forward in time, a behaviour also seen in Maxwell's equations, where so-called retarded waves advance in time while advanced waves, the other half of equally valid solutions which are usually ignored, propagate back in time. Why then, do physicists often ignore these solutions? This is the result of an assumption regarding the way events happen, an assumption about **causality**. Intuitively, the direction taken by events is tought to be the one going forward in time. This is due to the way we perceive consciousness and the world around us, how we assign meaning to it, if the mind is to be considered as a sense-making entity indeed (PPSISCO, 2025). But why attribute to reality a

characteristic result of our limits as conscious beings, when that is not the only available, sensible interpretation? Going back to the SE, if it's symmetrical when time-reversed, and it is, then it's also sensible to search for an interpretation that preserves that time symmetry even when considering collapse. Collapse, in an interpretation that allows time-symmetry, can be thought of as our perception as humans of the past and future boundary conditions of that quantum system intersecting and resolving themselves in what appears to be a present moment, in what instead is (subscribing to **Quantum Cosmology**) a whole universe and its wavefunction evolving in both the past and the future, a sort of "cosmic egg" whose ends are the ultimate past and future boundary conditions. Whatever these conditions are, the quantum system must satisfy both at once. A handshake between what the past gives and what the future needs, thats the nature of collapse in this theoretical framework. This dynamic allows for any quantum system to be considered time-reversible, eliminating the sudden, asymmetric collapse event, and instead attributing collapse too the nature of spacetime itself.

The evolution of this train of thought brings us to the **Transactional Interpretation of Quantum Mechanics**, by John Cramer (1986). According to TIQM, quantum systems send offer waves going forward in time, and confirmation waves, which travel backwards in time. A quantum event happens when a handshake across spacetime allows for both the offer wave and the confirmation wave's boundary conditions to coexist. Both the emitter, which produces the offer or retarded wave, and the absorber, which produces the confirmation or advanced wave, are thus now linked with a connection that spreads through both the future and the past.

Now that we understand how quantum mechanics are linked to microtubules, and can maybe infer how this would help solve classical, deterministic interpretations of Libet's finding, let us first explore if microtubules could indeed able to exhibit stable quantum effects between one another, and not only along one microtubule as discovered by Babcock et al. (2024). To do this, we provide a coarse, conceptual and in no way accurately representing the phenomenon, simulation of superradiance in the microtubules' tryptophan sites.

3.5 A conceptual simulation of superradiance

Superradiance, in the context of this thesis, is a phenomenon occurring whenever one or more quantum systems experience an enhancement in **radiation effects** (Dicke, 1954), i.e. the emission of energy through a medium in the form of waves or particles (Weisstein, 2014). The simulation we're about to present suggests that, even in semi-realistic biological disorder[2], both of the environment and the microtubules, the tryptophan networks are still able to achieve significant collective modes.

To achieve this result, the simulation comprises:

1. **Geometry** : The simulation doesn't use fully precise coordinates or atom-exact placements. Instead, it is a coarse, compromise-driven, structural representation of microtubules. According to the findings of Nogales et al. (1998) and Löwe et al. (2001), the simulation models microtubules as cylindrical lattices of 13 protofilaments of $\alpha-\beta$ tubulin dimers, each lattice with a 12.5nm radius. Each axial step (8nm distance between consecutive tubulin dimers) contains a singular tryptophan (from now on TRP) dipole, positioned in the lattice and oriented with idealized, non-realistic directions. Not a lifelike model then, rather a dipole network used to simulate and observe large-scale superradiance effects between many **chromophores**, which are the indole side-chain of the tryptophan, where there is an aromatic π electron system[3]. This kind of approximations are not unheard of in studies that explore collective radiative behaviour, as shown by the *Volta Key* of this whole thesis, the Babcock et al. (2024) article, in which researchers also employ a simplified model, focusing instead on the collective quantum effects. Their significant results shows that superradiance mostly depends on exactly what our model focuses on: between-dipole distances, environmental disorder and symmetry, rather than needing an extremely precise and computationally unfeasible atomistic model. This simulation is also built as to allow multiple microtubule segments (4 segments at maximum) to be placed at a center distance of circa 35nm, in layots such as: line; square; triangle.
2. **Physics:** Each tryptophan site is considered a dipole with **transition frequency wavenumber in medium[4]** equal to $\boldsymbol{k}=3.05*10^{7}m^{-1}$ [5], which is the value used in the

2 Which is the most common counter-argument for Penrose's and Hameroff's ORCH-OR (Tegmark, 2000)

3 A type of chemical bond where the π electrons are delocalized across multiple atoms, supporting the absorption of quanta of energy, the maintainance of excited states and dipole-dipole coupling (Kurian, 2017 & Babcock, 2024).

4 The specific frequency of EM radiation absorbed or emitted during the transition between ground and excited quantum states.

5 This value corresponds to circa 280nm.

Babcock et al. paper. The single-molecule **radiative rate**, i.e. the rate at which the molecule emits a quantum of energy (Lakowicz, 2006) also called a photon, is equal to $\gamma_0 = 1.15 * 10^9 s^{-¿}$. The refractive index of the medium has been set at $n = 1.36$, as to mimic a realistic cytosol environment for a general cell, as unfortunately neuronal cytosol refractive index values were not found (Curl et al., 2005). We then took the whole dipole-dipole interaction matrix, obtained via exact Maxwell Green's tensor and extracted the **effective non-Hermitian Hamiltonian** (James & Kerle, 2007): $H_{eff} = \Omega - i\frac{\Upsilon}{2}$, where **Ω** encodes coherent couplings[6] and **Y** encodes radiative decay (also called spontaneous emission, in this context). To get the superradiant eigenmodes, we used sparse eigensolvers, since we are dealing with a single-excitation manifold and would find mostly empty matrices. From that, we compute the **participation ratio**, i.e. the amount of tryptophan sites involved in the delocalization , finally obtaining the **quantum yield** for each mode with realistic non-radiative rate. This brief introduction serves as an index for the following, thourough explanation.

3. **The Model:** To even think about building a simulation regarding superradiance, we need to first find out at what rate the molecule (in our case a whole tryptophan molecule) decays radiatively, meaning at what rate the molecule emits a photon, going from its excited state to its ground state. The formula we have used is:

$$\gamma_0 = \frac{n\,\omega_0^3 \mu^2 L^2}{3\pi \epsilon_0 \hbar c_0^3}$$

In order, we define all symbols found in this equation.

- $\boldsymbol{\gamma_0}$: **Single-site radiative decay rate** (contrary to Y, which is instead the total rad. Decay)
- n : The refractive index of the medium, in this case and as previously mentioned it corresponds to the RI of any general cytosol, which is ~1.36.
- $\boldsymbol{\omega_0}$: the transition angular frequency, obtained via $\boldsymbol{\omega_0} = \frac{2\pi c_0}{\lambda_0}$
- λ_0 : this is the transition wavelength, taken directly form the Babcock et al. Paper, but in a vacuum. In our model, this corresponds exactly to 280nm.

6 Dipole-dipole couplings J_{ij} and site energies Δ_i, where $\boldsymbol{J_{ij}} \sim \frac{1}{r_{ij}^3}$

- μ : the tryptophan transition dipole moment's magnitude, which in our case has been set at 6 Debye (Mark, McKenna & Khajehpour, 2011).
- L : The Lorentz-Lorenz local field factor (Fick, 2001), $L=\frac{(\boldsymbol{n}^2+2)}{3}=1.28$. This factor (a rough approximation) is used to account for the differences between the macroscopic electric field and the stronger local field of the tryptophan molecule, itself sitting in a cavity surrounded by polarizable cytosol molecules. This value signifies a 28% stronger local field.
- ϵ_0 : The vacuum permittivity (https://docs.scipy.org/doc/scipy/reference/constants.html, last visited in 30/11/2025).
- $\hbar$: The reduced Planck's constant (https://physics.nist.gov/cgi-bin/cuu/Value?hbar, last visited in 30/11/2025).
- c_0 : The speed of light in a vacuum (Jackson, 1975).

After computing the single-site radiative decay rate accounting for L-L local field and the cytosol refraction index, the latter not present in the Babcock et al. paper, we turn to computing the dipole-dipole interactions. These interactions are governed by the **electromagnetic Green's tensor** (Martin & Paulus, 2001), which allows for the computation of the field radiated by dipole "*j*", evaluated at dipole "i" position, accounting for near, intermediate and far-field contributions. For each pair of tryptophan sites, their interaction is dictated by these parameters, needed to compute Green's tensor:

$\boldsymbol{r_{ij}}$: The distance from one site to another, as a vector.

$\hat{\mu}_i, \hat{\mu}_j$: Their orientation in their respective microtubules.

r_{ij}: Their separation distance, a scalar value.

$k=\frac{n\omega_0}{c_0}$: The wave number in the medium.

$\gamma_0=\frac{n\omega_0^3\mu^2L^2}{3\pi\epsilon_0\hbar c_0^3}$: The single molecule emission rate.

These parameters allow us to bring the exact Green's tensor into the equation:

$$\boldsymbol{G_{ij}} = \frac{e^{ikr_{ij}}}{4\pi r_{ij}} \left[\left(1+\frac{i}{\boldsymbol{kr_{ij}}}-\frac{1}{(\boldsymbol{kr_{ij}})^2}\right)(\hat{\mu}_i*\hat{\mu}_j)+\left(-1-\frac{3\boldsymbol{i}}{\boldsymbol{kr_{ij}}}+\frac{3}{(\boldsymbol{kr_{ij}})^2}\right)(\hat{\mu}_i*\hat{r}_{ij})(\hat{\mu}_j*\hat{r}_{ij})\right]$$

A more contextually friendly version of this equation: $\boldsymbol{G_{ij}} = J_{ij} - \frac{i}{2}\gamma_{ij}$, allows us to note that $J_{ij} = \Re(\boldsymbol{G_{ij}})$, the real part, is the coherent coupling term, encoding information about exciton transport, while the imaginary part contains $\gamma_{ij} = -2\Im(\boldsymbol{G_{ij}})$, which is the radiative coupling term and thus encodes information about superradiant behaviour. From these, we obtain the **Coherent Hamiltonian Matrix** and the **Collective Decay Matrix**, respectively:

$$\boldsymbol{\Omega} = \begin{pmatrix} \Delta_i & \boldsymbol{J_{ji}} \\ \boldsymbol{J_{ij}} & \Delta_j \end{pmatrix} \text{ and } \boldsymbol{Y_{ij}} = \begin{pmatrix} \gamma_0 & \gamma_{ji} \\ \gamma_{ij} & \gamma_0 \end{pmatrix}.$$

Here, the Δ_{ij} are the site energies with added disorder. It is now time to define the core of this simulation, i.e. the **Effective non-Hermitian Hamiltonian**:

$$H_{eff} = \Omega - i\frac{\Upsilon}{2}$$

This is the way the Effective Hamiltonian is defined in a single-excitation manifold such as the one we're using. More specifically, though in open quantum systems we would need to use a Lindblad equation, the fact that we are dealing with a single-exciton manifold, which is realistic for spontaneous UV emission in tryptophan, allows us to use this (relatively) simplified object. This procedure is standard for studies on quantum optics (Steck, 2007). We then diagonalize H_{eff} and solve $H_{eff}\boldsymbol{v_j} = \lambda_j \boldsymbol{v_j}$, getting the collective excitation mode as eigenvector $\mathbf{v_j}$, along with eigenvalue $\lambda_j = E_j - i\frac{\Gamma_j}{2}$. Here, the real part $E_j = \Re(\lambda_j)$ encodes information regarding the **collective energy**, while the imaginary part $\Gamma_j = -2\Im(\lambda_j)$ is the **collective decay rate of that mode.**

It is now possible to compare our resulting collective decay rate to γ_0, the single-site radiative decay, such that:

If $\Gamma_j \gg \gamma_0$, then we have what we have been searching for, the **superradiant mode**. If the ratio of $\frac{\Gamma_{max}}{\gamma_0} \gg 1$, then we have strong superradiance, meaning that finding the ratio to be grater than one tells us that **a single excitation is so delocalized across many tryptophan sites that it decays much faster than a single site** (Solano et al., 2017 ; Masson & Asenjo-Garcia, 2022)

If $\Gamma_j \sim \gamma_0$, then, of course, there is no significant difference between the single state decay rate

if $\Gamma_j \ll \gamma_0$, we have a **dark** or **subradiant** mode.

Regarding the eigenvector $\mathbf{v_i}$, we can extract:

v_i , the amplitude of the **brightest collective mode** of its site "I"

$|\boldsymbol{v_i}|^2$, which behaves similarly to a wavefunction Ψ, giving us the **probability weight** of that mode on its site "I", i.e. the probability that the excitation is on that specific chromophore.

But how delocalized exactly is the mode? We introduce the **Participation Ratio** (Gross & Haroche, 1982):

$$PR = \frac{1}{\sum_i |\boldsymbol{v_i}|^4}$$

Intuitively, the higher the ratio, the more significant the delocalization of the mode **inside a single microtubule**. To instead take a look at a multiple microtubule layout, we use $\text{w_per_mt} = \sum_{i \in MT_m} |\boldsymbol{v_i}|^2$, which tells us what quantity of the brightest mode (really the only mode we are interested in) is shared between MTs. If all MTs in the layout share similar weights, then we can talk about a **delocalized mode with non-trivial weight on multiple microtubules**. Last but not least, we are interested in what fraction of the tryptophan, chromophore sites actually emit a photon, instead of losing the energy non-radiatively. To obtain this information, we compute the **Quantum Yield**: $\boldsymbol{QY_j} = \frac{\Gamma_j}{\Gamma_j + \gamma_{nr}}$, relative of course to the brightest mode.

Now that the structure of the simulation has been clarified, along with the physics used in it, we can turn our gaze towards the results obtained.

4. RESULTS

We ran the simulation with different CLI arguments each time. Following this paragraph, a summary table for ease of access to the different characteristics of each run. For the results, median has been chosen as a preferred statistic due to the fact that superradiance is not linear and the mean would be exposed to unreasonable outliers producing very low or very high values due to subtle variations in disorder. For this reason, we will provide the 10th and 90th percentiles for PR, collective decay rate and QY.

Run Type	Objective	Added Disorder	Microtubules
Baseline	Ideal basis for comparison	None	1
Neuron-like environment	Provide a plausible result when accounting for biological disorder	σ=5e8, 3° tilt, 0.1nm jitter, 1% vacancy	1

Multi-MT Layout	Expand on known research to explore **inter-MT** coupling and superradiance effects	σ=5e8, 3° tilt, 0.1nm jitter, 1% vacancy	3 and 4

The results, respective to each run, are as follows:

For the baseline run, which is called results_baseline.jsonl:

$\frac{\Gamma_{max}}{\gamma_0} \approx 8.9$, ~ [8.88, 8.90]

$PR \gtrapprox 2000$, ~[1932, 2500]

$QY \approx 0.548$, ~ [0.547, 0.548]

We can clearly see that a perfect, ideal microtubular structure allows for strongly delocalized radiative modes, as seen in the Babcock et al. paper. This is a useful basis for comparison, as we can see that the theoretical maximum of this simulatiion has a collective radiative decay almost 9 times faster than the single molecule, while the percentage of energy actually emitted as photons is ~55%

For the first true run with realistic disorder, called results_neuronlike_sigma5e8.jsonl:

$\frac{\Gamma_{max}}{\gamma_0} \approx 7.1$, ~[6.7, 7.8]

$PR \approx 250 - 300$, ~[222, 480]

$QY \approx 0.49$ ~[0.47, 0.51]

Here, even after adding realistic disorder (positional, orientational, amplitude-related and 1% of vacancy of otherwise present tryptophan sites), we observe a delocalization across 300 sites, with the mode experiencing radiative decay 7 times faster than baseline γ_0. Of all the emitted energy, circa half of it is lost as emitted photons, the other is lost in a non-radiative manner. Both the collective radiative decay and the quantum yield are close to the results we would get from the previously presented theoretical maximum, indicating that **even with realistic cytosolic disorder, microtubules seem to engage in strongly delocalized modes, experiencing a significant superradiant effect (a 7-fold increase).**

For the Multi-MT triangular layout, called results_3mt_tri_35nm.jsonl:

$\frac{\Gamma_{max}}{\gamma_0} \approx 3.92$, ~ [3.71, 4.57]

$PR \approx 442$, [346, 637]

$QY \approx 0.34$, ~ [0.33, 0.38]

inter-MT delocalization: ~[0.33, 0.54, 0.12]

In this multi-MT triangular layout, we can conclude that , though the radiative decay across microtubules is not as enhanced as we have seen with single microtubule segments, we still get a 4-fold increase in the rate of emission, **with a higher PR compared to the realistic, single-MT segments.** This superradiant mode experiences a ~20% reduction in Quantum Yield compared to single-MT, and it's delocalized mostly across the first two MTs. This particular run supports the idea that **superradiant enhancement is observable even across multiple microtubules**, which is an assumption not experimentally supported in the Babcock et al. paper.

For the multi-MT square layout, called results_4mt_square_35nm.jsonl :

$\frac{\Gamma_{max}}{\gamma_0} \approx 7.72$, [7.40, 7.87]

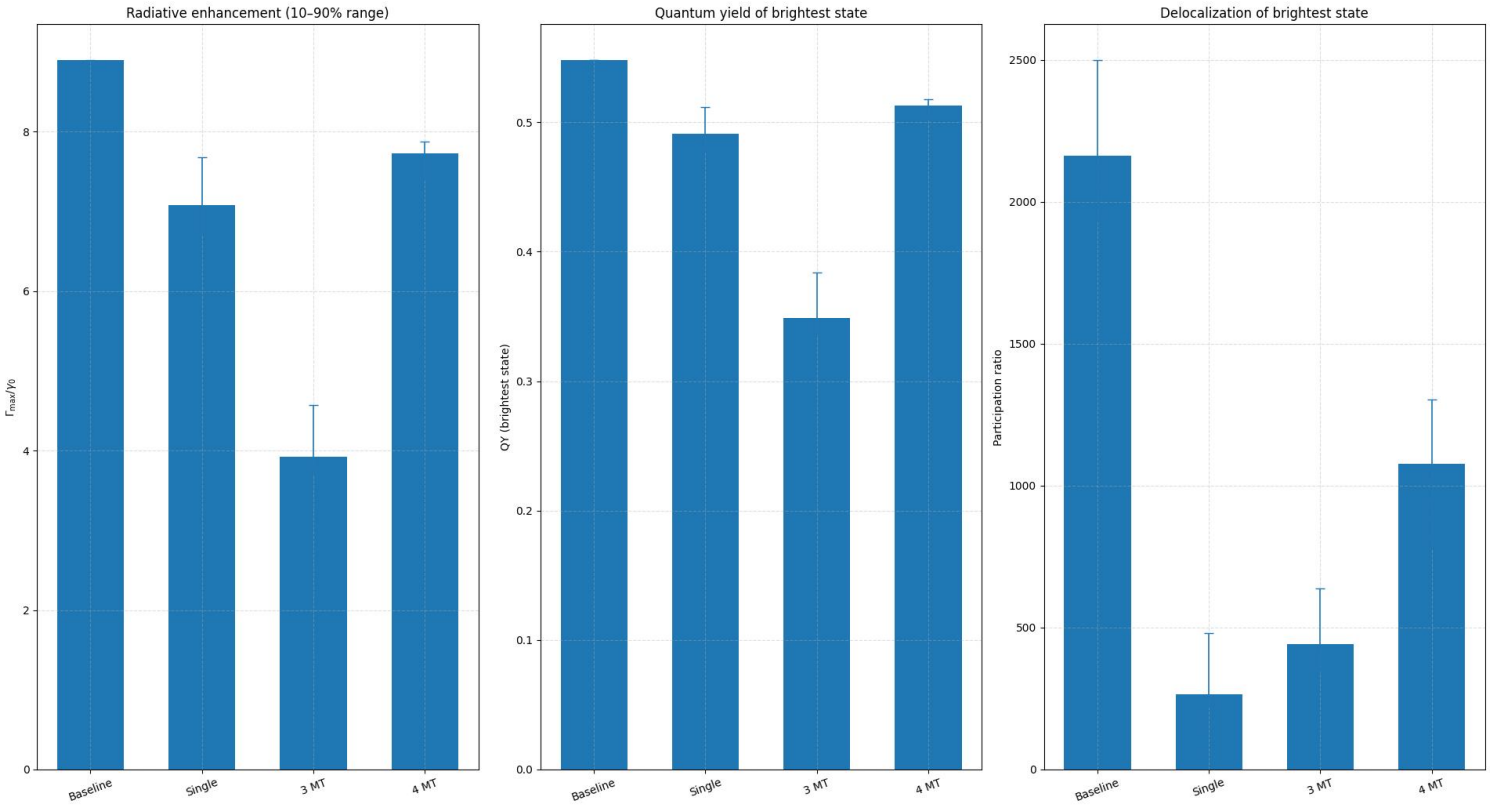
$PR \approx 1078$, [782, 1303]

$QY \approx 0.513$, [0.503, 0.518]

inter-MT delocalization: ~ [0.42, 0.04, 0.06, 0.47]

The most interesting results as of yet, we can observe that when a square layout of 4 MTs is selected, we reach a collective decay rate comparable to that of the single MT run, which is also comparable to the ideal baseline. The PR skyrockets to circa 1000 sites involved, though the exciton delocalization seems to be spread mostly between 2 of the 4 MTs. **This findings suggest that more numerous groups of relatively close toghether microtubules with the correct orientation (here we see a parallel layout) might exhibit more intense superradiant effects, spread mostly between 2 of the many and concerning a large portion of the tryptophan sites.**

These results suffer no particular bias due to the seed. Here are the plotted results in form of bar graphs:



These plots show that when 3 microtubules are placed in a triangular layout, we observe worse results overall, even compared with the single MT run. That changes completely when using a square layout, though: **the 4 MTs run shows the best results**. Since the number of tryptophan sites increases from the single MT run to the 3 MTs run, we can assume that this is not due to the increase of sites *per se*, rather **their parallel configuration might potentiate superradiant effects**, with values approaching our ideal baseline.

5. LIMITATIONS AND FURTHER RESEARCH

Technical limitations forbade running the simulation with more microtubules, limiting the reach of the shown results and following conclusion. The simulation is nonetheless satisfactory in its conceptual, coarse-grained role, providing a reasonable basis for disorder-including superradiance models. The glaring weakness of the whole discourse is the lack of hard proofs, with a still growing *corpus* of experiments regarding the role of microtubules as sub-neural information processing hubs. When we turn our heads to TIQM and Quantum Cosmology, we do reach these conclusions starting from a mathematical quality of the SE, i.e. the fact that it produces valid values even when

time-reversed. Providing proof or support for these theories, though, is all but trivial. As of now we have no way of measuring or interacting with moments of time different than the present, and this suggests that for the known time we might have to be content with mathematical proofs or hints. While respectable and eye-opening, such theories might need more experimental proofs if they are to be widely accepted, by both academia and the public. As of now, we can state that the universe is non-local and non-real(Aspect, Clauser, & Zeilinger, 2022), hoping further research brings light on our related phenomena and speculations.

6. CONCLUSIONS

The results we've gathered from this simulation suggest that not only superradiance is possible across microtubules, but that when placed in parallel, this effect is even stronger. These structures have been proposed as potential candidates for sub-neuronal processing, a claim supported by anesthetic research and its reviews (Wiest, 2025), but also by recent advancements in the study of neurodegenerative diseases (Fernandez-Venezuela et al., 2020). Here we propose that superradiance, or analogue quantum-related processes like FRET (Li et al., 2024), therefore excitonic or vibrational processes, might be a candidate for explaining how this sub-neural informational processes happen. This admittedly speculative conclusion does and will require extensive further research, but when paired with other hard-to-verify theories such as TIQM and Quantum Cosmology, it leads us to an interesting, paradigm-breaking conclusion. Assuming that microtubules are the seat of consciousness (yet to be proven), that superradiance or *simila* are the mechanism of action (also yet to be proven), and that quantum systems are defined by boundary conditions both in the past and the future as per TIQM, we can reach an exhilarating statement.

Libet's findings and their deterministic interpretations, which rob humanity of a free will that might have never been there, are not certain anymore. If the universe itself is a tetra-dimensional block where past, present and future coexist and act (Carroll, 2010), then in a non-retrocausal manner, both the future and the past contribute to what happens in our perceived present. This leads the author of this thesis to state, with potentially misplaced confidence, that **it matters not whether the Readiness Potential comes before the thought, as both the RP and the "following" thought**

exist in the same spacetime, both defined and influenced by their boundary conditions, both cooperating in their creation.

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APPENDIX

Here the python code for the simulations is provided as plain text:

- geometry.py:

```
# geometry.py
# frankenstein monster form stack exchange, youtube and AI. touch nothing or it breaks lol
import numpy as np

def _unit(v):
v = np.asarray(v, float)
n = np.linalg.norm(v, axis=-1, keepdims=True)
return v / np.clip(n, 1e-30, None)

def _random_unit_vectors(n, rng):
a = rng.normal(size=(n, 3))
a /= np.linalg.norm(a, axis=1, keepdims=True)
return a

# straight protofilament 1d line along z
def protofilament(n_sites: int, spacing: float = 8.0e-9, mu_dir=(1,0,0)):
"""

"""
z = np.arange(n_sites, dtype=float) * spacing
R = np.c_[np.zeros_like(z), np.zeros_like(z), z] # (N,3)
MU = np.tile(_unit(mu_dir), (n_sites,1)) # (N,3)
return R, MU

# ring on xy-plane
def ring(n_sites: int, radius: float, orientation: str = "tangential"):

phi = 2*np.pi*np.arange(n_sites)/n_sites
x, y = radius*np.cos(phi), radius*np.sin(phi)
R = np.c_[x, y, np.zeros_like(x)]

if orientation == "tangential":
MU = np.c_[-np.sin(phi), np.cos(phi), np.zeros_like(phi)]
elif orientation == "radial":
MU = np.c_[np.cos(phi), np.sin(phi), np.zeros_like(phi)]
elif orientation == "z":
MU = np.c_[np.zeros_like(phi), np.zeros_like(phi), np.ones_like(phi)]
else:
raise ValueError("orientation must be tangential|radial|z")
```

```
MU = _unit(MU)
return R, MU

# 13-proto microtubule (simple helical scaffold)
def microtubule_13proto(n_axial: int,
a_axial: float = 8.0e-9,
radius: float = 12.5e-9,
helix_pitch: float = 3*8.0e-9,
orientation: str = "tangential",
phi0: float = 0.0,
dz: float = 0.0):
"""
"""
P = 13
k_idx = np.arange(n_axial, dtype=float)
p_idx = np.arange(P, dtype=float)

# Mesh over (protofilament p, axial index k)
k_grid, p_grid = np.meshgrid(k_idx, p_idx, indexing="ij")
dphi = 2*np.pi / P if helix_pitch is None else 2*np.pi*(a_axial/helix_pitch)


phi = 2*np.pi*(p_grid / P) + k_grid * dphi + phi0 # helical advance
z = k_grid * a_axial + dz
x = radius*np.cos(phi)
y = radius*np.sin(phi)

R = np.stack([x, y, z], axis=-1).reshape(-1, 3)

if orientation == "tangential":
MU = np.stack([-np.sin(phi), np.cos(phi), np.zeros_like(phi)], axis=-1)
elif orientation == "radial":
MU = np.stack([np.cos(phi), np.sin(phi), np.zeros_like(phi)], axis=-1)
elif orientation == "axial":
MU = np.stack([np.zeros_like(phi), np.zeros_like(phi), np.ones_like(phi)], axis=-1)
else:
raise ValueError("orientation must be tangential|radial|axial")

MU = _unit(MU.reshape(-1,3))
return R, MU


def multi_microtubules(n_mt: int,
n_axial: int,
a_axial: float = 8.0e-9,
radius: float = 12.5e-9,
```

```
orientation: str = "tangential",
layout: str = "line",
center_spacing: float = 35e-9,
rng: np.random.Generator = None):
"""
"""
assert n_mt >= 1
if rng is None:
rng = np.random.default_rng()

# Place MT centers in xy-plane
if layout == "line":
xs = (np.arange(n_mt) - (n_mt-1)/2.0) * center_spacing
centers = np.c_[xs, np.zeros_like(xs), np.zeros_like(xs)]
elif layout == "triangle":
assert n_mt == 3, "triangle layout requires n_mt=3"
s = center_spacing
centers = np.array([
[0.0, +s/np.sqrt(3), 0.0],
[-s/2.0, -s/(2*np.sqrt(3)), 0.0],
[+s/2.0, -s/(2*np.sqrt(3)), 0.0],
])
elif layout == "square":
assert n_mt == 4, "square layout requires n_mt=4"
s = center_spacing / np.sqrt(2) # so nearest-neighbor distance is approximately center_spacing
centers = np.array([
[-s, -s, 0.0],
[-s, +s, 0.0],
[+s, -s, 0.0],
[+s, +s, 0.0],
])
else:
raise ValueError("layout must be: line | triangle | square")

R_all = []
MU_all = []
labels = []

for m in range(n_mt):
phi0 = float(rng.uniform(0.0, 2.0*np.pi))
dz = float(rng.uniform(0.0, a_axial))
R, MU = microtubule_13proto(n_axial=n_axial, a_axial=a_axial,
radius=radius, helix_pitch=None,
orientation=orientation, phi0=phi0, dz=dz)
# translate by center
R = R + centers[m][None, :]
```

```
R_all.append(R)
MU_all.append(MU)
labels.append(np.full(R.shape[0], m, dtype=np.int32))

R_out = np.vstack(R_all)
MU_out = np.vstack(MU_all)
labels = np.concatenate(labels)
return R_out, MU_out, labels
```

- Green's tensor and H_{eff}:

```
# physics_hook_fast.py
from __future__ import annotations
import numpy as np
import math
from typing import Tuple, Dict, Any
from scipy.spatial import cKDTree
from scipy.sparse import csr_matrix, diags
from scipy.sparse.linalg import eigs

# optional numba kernel, set NUMBA=1 to use. on my machine i didn't manage to make it work
tbh
NUMBA = 0
if NUMBA:
from numba import njit

def _omega_gamma_pairs(mu_i, mu_j, rvec, k, gamma0):
"""Vectorized Lehmberg for arrays of pairs.
mu_i, mu_j: (M,3); rvec: (M,3); returns (Omega, Gamma) shape (M,)"""
r = np.linalg.norm(rvec, axis=1)
x = k * r
rhat = rvec / np.clip(r[:, None], 1e-30, None)

ui_uj = np.einsum('ij,ij->i', mu_i, mu_j)
ui_r = np.einsum('ij,ij->i', mu_i, rhat)
uj_r = np.einsum('ij,ij->i', mu_j, rhat)

A = ui_uj - ui_r * uj_r
B = ui_uj - 3.0 * ui_r * uj_r

# safe sin/cos
small = (x < 1e-6)
sinx = np.empty_like(x)
cosx = np.empty_like(x)
sinx[small] = x[small] - (x[small]**3)/6.0
```

```
cosx[small] = 1.0 - 0.5*(x[small]**2)
sinx[~small] = np.sin(x[~small])
cosx[~small] = np.cos(x[~small])

invx = np.where(x!=0, 1.0/x, 0.0)
invx2 = invx*invx
invx3 = invx2*invx

# Γ_ij
Gamma = (1.5*gamma0) * ( A*(sinx*invx) + B*(cosx*invx2 - sinx*invx3) )
# Ω_ij
Omega = -(0.75*gamma0) * ( A*(cosx*invx) - B*(sinx*invx2 + cosx*invx3) )
return Omega, Gamma

if NUMBA:
_omega_gamma_pairs = njit(_omega_gamma_pairs) # type: ignore

def build_sparse_mats(R: np.ndarray, MU: np.ndarray, k: float, gamma0: float,
r_cut: float) -> Tuple[csr_matrix, csr_matrix]:
"""
R: (N,3), MU: (N,3) unit dipoles
returns Ω, Y as CSR (with Y_ii = γ0)
"""
N = R.shape[0]
tree = cKDTree(R)
pairs = tree.query_pairs(r_cut, output_type='ndarray') # (M,2)
if pairs.size == 0:
Omega = diags(np.zeros(N), format='csr')
Upsilon = diags(np.full(N, gamma0), format='csr')
return Omega, Upsilon

i = pairs[:,0]; j = pairs[:,1]
rvec = R[j] - R[i]
Om, Gm = _omega_gamma_pairs(MU[i], MU[j], rvec, k, gamma0)

# assemble symmetric sparse (only off-diagonals here)
data_O = np.concatenate([Om, Om])
rows_O = np.concatenate([i, j])
cols_O = np.concatenate([j, i])

data_G = np.concatenate([Gm, Gm])
rows_G = rows_O
cols_G = cols_O

Omega = csr_matrix((data_O, (rows_O, cols_O)), shape=(N, N))
Upsilon = csr_matrix((data_G, (rows_G, cols_G)), shape=(N, N))
```

```
# add self-decay γ0 on diagonal
Upsilon = Upsilon + diags(np.full(N, gamma0), format='csr')
return Omega, Upsilon

def heff_from_ou(Omega: csr_matrix, Upsilon: csr_matrix) -> csr_matrix:
return Omega.astype(np.complex128) - 0.5j * Upsilon.astype(np.complex128)

def brightest_modes(Heff: csr_matrix, k_eigs: int = 8) -> Tuple[np.ndarray, np.ndarray]:
vals, vecs = eigs(Heff, k=k_eigs, which="LM")
return vals, vecs

def summarize_brightness(evals: np.ndarray, gamma0: float, top_n: int = 3):
gammas = (-2.0 * np.imag(evals)).real
order = np.argsort(gammas)[::-1]
top = order[:top_n]
return {
"Gamma_over_gamma0_top": (gammas[top] / gamma0).tolist(),
"Gamma_top": gammas[top].tolist(),
"evals_top": evals[top].tolist(),
"Gamma_max_over_gamma0": float(np.max(gammas) / gamma0),
}
```

- physics consants (constants.py, name might change in the code so check before running):

```
# constants.py DON'T USE PAPER MODE, USE NEURONLIKE

import math
from scipy import constants as cst
# fundamental constants
c0 = 299_792_458.0 # m/s, speed of causality
eps0 = cst.epsilon_0 # F/m
hbar = 1.054_571_817e-34 # J·s reduced h

# Molecular line (vacuum)
lam0 = 280e-9 # central wavelength in vacuum of tryptophan, as per papaer [m]
mu_D = 6.0 # transition dipole magnitude [Debye] (paper uses ~6 D)
mu = mu_D * 3.33564e-30 # C·m
omega0 = 2.0 * math.pi * c0 / lam0 # angular frequency in vacuum [s^-1]

# Environment (neuron-ish if paper is silent). DO NOT USE PAPER MODE
n_med = 1.36 # neuronal cytosol/axon ~1.35–1.38
use_local_field = True # toggle Lorentz–Lorenz only in neuron mode

# --- Mode switch ---
paper_mode = False # True => reproduce paper's convention

# distance r cutoff
```

```
r_cut = 300e-9
def gamma0_from_paper():
"""
Paper reports a single-site linewidth ~2.73e-3 cm^-1 (gives τ ≈ 1.9 ns).
Convert wavenumber linewidth [cm^-1] -> angular rate [s^-1] via 2π c.
"""
gamma0_cm1 = 2.73e-3
return gamma0_cm1 * 2.0 * math.pi * (c0 * 100.0)

# k and gamma0 according to mode
if paper_mode:
k = 2.0 * math.pi / lam0 # vacuum wave number k0 [m^-1]
gamma0 = gamma0_from_paper() # single-site rate [s^-1]
else:
k = (n_med * omega0) / c0 # medium wave number [m^-1]
L = ((n_med**2 + 2.0) / 3.0) if use_local_field else 1.0
gamma0 = (n_med * omega0**3 * (mu**2) * (L**2)) / (3.0 * math.pi * eps0 * hbar * c0**3)

# converters (so the terminal produces intellegible output)
J_to_eV = 1.0 / 1.602_176_634e-19
s_to_ps = 1e-12

# --- Quantum yield and gamma ---
#QY_single = 0.12 # tryptophan in water-like environment
#gamma_nr = C.gamma0 * (1.0 - QY_single) / QY_single # ≈ 7.33 * gamma0
#THESE ONES WE PASS IN ARGS WHEN CLI
```

- main file (run_segments_physics.py):

```
# run_segments_physics.py

# this too an amalgamation of many different sources, cited of course in the thesis
import importlib, os, json, numpy as np
from multiprocessing import Pool, cpu_count
from scipy.sparse import diags

G = importlib.import_module("geometry")
PH = importlib.import_module("physics_hook_fast") # vectorized/numba-optional (numba is much
faster but good luck)
C = importlib.import_module("SIM_constants") # must define k, gamma0, r_cut
try:
print(f"[constants] n={getattr(C,'n_med',None)} L={getattr(C,'L',1.0):.3f} "
f"k={C.k:.3e} 1/m gamma0={C.gamma0:.3e} s^-1")
except Exception:
pass

import argparse
```

```
def parse_args():
p = argparse.ArgumentParser()
p.add_argument("--segments", type=int, default=30, help="number of segments")
p.add_argument("--slice_um", type=float, default=3.0, help="slice length in microns")
p.add_argument("--r_cut_nm", type=float, default=300.0, help="neighbor cutoff in nm")
p.add_argument("--k_eigs", type=int, default=8, help="# of brightest modes to compute")
p.add_argument("--orientation", type=str, default="tangential",
choices=["tangential","radial","axial"])
p.add_argument("--sigma", type=float, default=0.0,
help="static diagonal disorder std (s^-1); 0 disables")
# inside parse_args() in run_segments_physics.py
p.add_argument("--orient_sigma_deg", type=float, default=0.0,
help="per-site dipole tilt std dev (degrees)")
p.add_argument("--pos_sigma_nm", type=float, default=0.0,
help="per-site positional jitter std dev (nm)")
p.add_argument("--vacancy", type=float, default=0.0,
help="fraction of sites randomly removed [0..1]")

p.add_argument("--qy_single", type=float, default=0.12,
help="single-Trp quantum yield")
p.add_argument("--outfile", type=str, default="segments_results.jsonl",
help="results path (JSONL append)")
p.add_argument("--seed", type=int, default=12345)
p.add_argument("--n_mt", type=int, default=1,
help="number of parallel microtubules in a segment (1,2,3,4)")
p.add_argument("--mt_layout", type=str, default="line",
choices=["line","triangle","square"],
help="layout of MT centers for n_mt>1")
p.add_argument("--center_spacing_nm", type=float, default=35.0,
help="center-to-center spacing between MT axes (nm)")
p.add_argument("--workers", type=int, default=max(1, cpu_count()-1),
help="number of parallel worker processes")


return p.parse_args()

ARGS = parse_args()

#knobs (from CLI args)
L_SLICE_M = ARGS.slice_um * 1e-6
N_SEGMENTS = ARGS.segments
ORIENTATION = ARGS.orientation
A_AXIAL = 8.0e-9 # dimer spacing stays fixed
K_EIGS = ARGS.k_eigs
RESULTS_PATH = ARGS.outfile
```

```
RNG_SEED = ARGS.seed

# QY and non-radiative rate
QY_SINGLE = ARGS.qy_single
GAMMA_NR = C.gamma0 * (1.0 - QY_SINGLE) / QY_SINGLE # ≈ 7.33*gamma0 if QY_SINGLE=0.12

# Override cutoff in constants (so physics uses the CLI cutoff) DO NOT CHANGE OTHERWISE CLI
DOESN'T LET YOU SET PARAMETERS
C.r_cut = ARGS.r_cut_nm * 1e-9


def n_axial_from_length(Lm, a_ax=A_AXIAL):
n = int(max(1, np.floor(Lm / a_ax) + 1))
return n

def participation_ratio(v):
v = v / np.linalg.norm(v)
return float(1.0 / np.sum(np.abs(v)**4))

def _random_unit_vectors(n, rng):
a = rng.normal(size=(n, 3))
a /= np.linalg.norm(a, axis=1, keepdims=True)
return a

def _rodrigues_batch(v, axes, angles):
"""
Rotate each row of v by angle[i] around axes[i] (all (N,3)).
v_rot = v*cosθ + (k×v)*sinθ + k*(k·v)*(1−cosθ)
Entirely stolen from a stack exchange post i'm unable to recover
"""
c = np.cos(angles)[:, None]
s = np.sin(angles)[:, None]
kv = np.sum(axes * v, axis=1, keepdims=True) # (N,1)
k_cross_v = np.cross(axes, v) # (N,3)
return v * c + k_cross_v * s + axes * kv * (1.0 - c)


def build_one_segment(rng: np.random.Generator):
n_ax = n_axial_from_length(L_SLICE_M, A_AXIAL)

if ARGS.n_mt == 1:
# single MT (your old code)
phi0 = float(rng.uniform(0.0, 2.0*np.pi))
dz = float(rng.uniform(0.0, A_AXIAL))
R, MU = G.microtubule_13proto(
```

```
n_axial=n_ax, a_axial=A_AXIAL,
radius=12.5e-9, helix_pitch=None,
orientation=ORIENTATION, phi0=phi0, dz=dz
)
labels = np.zeros(R.shape[0], dtype=np.int32)
else:
# multi-MT slice
R, MU, labels = G.multi_microtubules(
n_mt=ARGS.n_mt, n_axial=n_ax, a_axial=A_AXIAL,
radius=12.5e-9, orientation=ORIENTATION,
layout=ARGS.mt_layout,
center_spacing=ARGS.center_spacing_nm * 1e-9,
rng=rng
)
phi0, dz = None, None # not meaningful for multi-MT

# APPLY STRUCTURAL IMPERFECTIONS, DISORDER ETC ETC
if ARGS.pos_sigma_nm > 0.0:
R = R + rng.normal(0.0, ARGS.pos_sigma_nm * 1e-9, size=R.shape)

if ARGS.orient_sigma_deg > 0.0:
N = MU.shape[0]
axes = _random_unit_vectors(N, rng)
angles = rng.normal(0.0, np.deg2rad(ARGS.orient_sigma_deg), size=N)
MU = _rodrigues_batch(MU, axes, angles)
MU /= np.linalg.norm(MU, axis=1, keepdims=True)

if ARGS.vacancy > 0.0:
keep = rng.random(R.shape[0]) > ARGS.vacancy
if not np.any(keep):
keep[rng.integers(0, R.shape[0])] = True
R, MU, labels = R[keep], MU[keep], labels[keep]
# END DISORDER
return dict(R=R, MU=MU, labels=labels, phi0=phi0, dz=dz, n_axial=n_ax)

def run_one(idx_seg_and_seg):
idx, seg = idx_seg_and_seg
# per-worker RNG (reproducible and independent across processes)
rng = np.random.default_rng((1234567, idx))

# disorder strength (standard deviation) in s^-1 (from CLI)
sigma = ARGS.sigma

R, MU = seg["R"], seg["MU"]
N = R.shape[0]
```

```
# Build Ω, Y
Omega, Upsilon = PH.build_sparse_mats(R, MU, C.k, C.gamma0, C.r_cut)
Omega = Omega.astype(np.float32)
Upsilon = Upsilon.astype(np.float32)

# Add static diagonal disorder (site energies) to Ω for realism:
if sigma > 0.0:
delta = rng.normal(loc=0.0, scale=sigma, size=N)
Omega = Omega + diags(delta, offsets=0, format="csr")

# H_eff = Ω - i Y/2
Heff = PH.heff_from_ou(Omega, Upsilon)

# Brightest modes
vals, vecs = PH.brightest_modes(Heff, k_eigs=K_EIGS)

# Radiative rates Γ_j from eigenvalues (compute ONCE)
gammas = (-2.0*np.imag(vals)).real
bidx = int(np.argmax(gammas))
Gamma_bright = float(gammas[bidx])

# Quantum yield for brightest mode and for all returned modes
QY_bright = Gamma_bright / (Gamma_bright + GAMMA_NR)
order = np.argsort(gammas)[::-1]
QY_all = (gammas[order] / (gammas[order] + GAMMA_NR)).astype(float).tolist()

# Summaries (uses vals to get Γ/γ0 ratios for top modes)
summ = PH.summarize_brightness(vals, C.gamma0, top_n=min(3, K_EIGS))

# Participation ratio of the brightest state
labels = seg["labels"]
v = vecs[:, bidx]
v = v / np.linalg.norm(v)
PR = float(1.0 / np.sum(np.abs(v)**4))

# weight on each MT (sum of |v|^2 over sites belonging to that MT)
n_mt_present = int(labels.max()) + 1
w_per_mt = []
for m in range(n_mt_present):
mask = (labels == m)
w_per_mt.append(float(np.sum(np.abs(v[mask])**2)))

row = {
"idx": idx,
"n_axial": int(seg["n_axial"]),
"N_sites": int(N),
```

```
"phi0": seg["phi0"],
"dz_m": seg["dz"],
"Gamma_max_over_gamma0": float(summ["Gamma_max_over_gamma0"]),
"Gamma_over_gamma0_top": [float(x) for x in summ["Gamma_over_gamma0_top"]],
"PR_brightest": float(PR),
"sigma_disorder_s^-1": float(sigma),
# QY & rates
"Gamma_bright": Gamma_bright,
"QY_bright": float(QY_bright),
"gamma_nr": float(GAMMA_NR),
"QY_single_assumed": float(QY_SINGLE),
"QY_top": QY_all[:min(3, len(QY_all))],
# multi microtubule info REMEMBER DONT TRY 5 MTS OR MACHINE CRASHES
"n_mt": int(ARGS.n_mt),
"mt_layout": ARGS.mt_layout,
"center_spacing_nm": float(ARGS.center_spacing_nm),
"bright_weights_per_mt": w_per_mt
}

# Save run configuration
row.update({
"config": {
"slice_um": ARGS.slice_um,
"r_cut_nm": ARGS.r_cut_nm,
"k_eigs": ARGS.k_eigs,
"orientation": ARGS.orientation,
"sigma": ARGS.sigma,
"qy_single": ARGS.qy_single,
"orient_sigma_deg": ARGS.orient_sigma_deg,
"pos_sigma_nm": ARGS.pos_sigma_nm,
"vacancy": ARGS.vacancy,
"n_mt": int(ARGS.n_mt),
"mt_layout": ARGS.mt_layout,
"center_spacing_nm": float(ARGS.center_spacing_nm),
}
})


return row

def save_row(path, row):
with open(path, "a") as f:
f.write(json.dumps(row) + "\n")

def main():
rng = np.random.default_rng(RNG_SEED)
```

```
segments = [build_one_segment(rng) for _ in range(N_SEGMENTS)]

# resume-safe
done = set()
if os.path.exists(RESULTS_PATH):
with open(RESULTS_PATH, "r") as f:
for line in f:
try:
obj = json.loads(line)
done.add(int(obj["idx"]))
except Exception:
pass

work = [(i, seg) for i, seg in enumerate(segments) if i not in done]
print(f"Slice length: {L_SLICE_M*1e6:.2f} µm | n_axial={segments[0]['n_axial']} |
sites/MT={segments[0]['R'].shape[0]}")
print(f"Segments to run: {len(work)} (skipping {len(segments)-len(work)} already done)")
print(f"Cutoff: {ARGS.r_cut_nm:.0f} nm | k_eigs={K_EIGS} | orientation={ORIENTATION} | "
f"sigma={ARGS.sigma:.2e} s^-1 | QY_single={QY_SINGLE:.2f}"
f"tiltσ={ARGS.orient_sigma_deg:.1f}° | posσ={ARGS.pos_sigma_nm:.2f} nm |
vac={ARGS.vacancy:.2f}")
print(f"MTs: n_mt={ARGS.n_mt}, layout={ARGS.mt_layout},
center_spacing={ARGS.center_spacing_nm:.1f} nm")


# parallel + progress
n_workers = ARGS.workers
try:
from tqdm import tqdm
iterator = tqdm
except Exception:
iterator = lambda x, **k: x

with Pool(processes=n_workers) as pool:
for row in iterator(pool.imap_unordered(run_one, work, chunksize=1),
total=len(work), desc="Segments"):
save_row(RESULTS_PATH, row)

# aggregate
data = []
with open(RESULTS_PATH, "r") as f:
for line in f:
try:
data.append(json.loads(line))
except Exception:
pass
```

```
if not data:
print("\n== SUMMARY ==\n(no results found — did the run finish?)")
return

g = np.array([d["Gamma_max_over_gamma0"] for d in data], float)
pr = np.array([d["PR_brightest"] for d in data], float)
Ns = np.array([d["N_sites"] for d in data], int)

def q(a, p): return float(np.quantile(a, p))

print("\n== SUMMARY ==")
print("segments:", len(data))
print("sites/segment median [p10, p90]:", int(np.median(Ns)),
f"[{int(q(Ns,0.10))}, {int(q(Ns,0.90))}]")
print("Γ_max/γ0 median [p10, p90] mean:",
f"{np.median(g):.2f}",
f"[{q(g,0.10):.2f}, {q(g,0.90):.2f}]",
f"{g.mean():.2f}")
print("PR_bright median [p10, p90] mean:",
f"{np.median(pr):.1f}",
f"[{q(pr,0.10):.1f}, {q(pr,0.90):.1f}]",
f"{pr.mean():.1f}")

# QY summary (only if present in rows)
if "QY_bright" in data[0]:
qy = np.array([d["QY_bright"] for d in data], float)
print("QY_bright median [p10, p90] mean:",
f"{np.median(qy):.3f}",
f"[{q(qy,0.10):.3f}, {q(qy,0.90):.3f}]",
f"{qy.mean():.3f}")
else:
print("QY_bright: (not available — did you add QY fields to each row in run_one?)")

if __name__ == "__main__":
main()
```